Forecasting Keyword Trends with Prediction Intervals:
Enhancing Decision-Making Under Uncertainty

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1. ***Abstract*—**

This paper explores the application of machine learning to forecast keyword trends using data sourced from Google Trends, aiming to provide data-driven support for keyword selection. However, keyword trend data inherently involves aleatoric uncertainty (random noise) and epistemic uncertainty (model uncertainty), often overlooked in traditional models. When keyword difficulty is similar between options, relying solely on direct predictions from machine learning models can be misleading. This study proposes the integration of prediction intervals into forecasting models, offering a quantifiable measure of the model's confidence. Wider intervals signal lower model certainty and highlight potential volatility in trend predictions. By incorporating prediction intervals, we advocate for a more transparent and responsible AI approach, empowering SEO professionals and digital marketers to make more informed decisions grounded not just in forecasts, but in the reliability of those forecasts.

2. Introduction

In the realm of digital marketing and search engine optimization (SEO), understanding and anticipating keyword trends is pivotal for crafting effective content strategies. Tools like Google Trends offer valuable insights into the relative popularity of search terms over time, enabling marketers to identify emerging topics and allocate resources accordingly. However, traditional forecasting models often provide point estimates without conveying the associated uncertainty, potentially leading to overconfident decisions and suboptimal outcomes. Keyword trend data inherently encompasses two types of uncertainty: aleatoric uncertainty, stemming from the inherent randomness and noise in user search behavior, and epistemic uncertainty, arising from limitations in the predictive models themselves. Neglecting these uncertainties can result in misleading forecasts, especially when the predicted trends of different keywords are closely aligned. For instance, when choosing between two keywords with similar forecasted popularity, relying solely on point predictions may obscure the underlying risks associated with each option.

To address this challenge, our study explores the integration of prediction intervals into keyword trend forecasting models. Prediction intervals provide a range within which future observations are expected to fall, offering a quantifiable measure of the model's confidence. By incorporating these intervals, we aim to enhance the transparency and reliability of forecasts, enabling marketers and automated decision-making systems to make more informed choices. We applied three distinct models to forecast trends for the keywords "Book" and "Football" using five years of Google Trends data:

- The Auxiliary Interval Predictor from IBM's Uncertainty Quantification 360 toolkit.
- Gradient Boosting Regressor (GBR) configured to predict the 5th and 95th percentiles.
- Random Forest (RF) model estimating the 90% prediction interval.

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Our evaluation focused on two key metrics: Prediction Interval Coverage Probability (PICP), which assesses the proportion of actual observations falling within the predicted intervals, and Mean Prediction Interval Width (MPIW), indicating the average width of these intervals. The results revealed that while "Football" exhibited a higher average predicted rank, it also displayed lower PICP and wider MPIW, suggesting greater volatility and uncertainty. Conversely, "Book" demonstrated higher coverage with narrower intervals, indicating more stable and reliable forecasts.

These findings underscore the importance of incorporating uncertainty quantification into keyword trend forecasting. By moving beyond point estimates and embracing prediction intervals, marketers and AI-driven systems can better assess the risks associated with different keywords, leading to more robust and informed decision-making processes.

3. Literature Review

The domain of Search Engine Optimization (SEO) has experienced significant transformation over the years, driven largely by the rapid expansion of web content and the evolving behaviors of users in the digital environment. In its early stages, SEO efforts predominantly emphasized straightforward tactics such as keyword frequency and the accumulation of backlinks. However, contemporary SEO practices are now shaped by the intricate and often opaque nature of search engine ranking algorithms. This paper contributes to a growing body of research focused on analyzing and optimizing SEO strategies through analytical and computational models, particularly in scenarios marked by uncertainty. A central objective of this work is to bridge the gap between existing theoretical models and the practical realities faced by webmasters who operate without full knowledge of the mechanisms that determine search rankings.

Previous studies have employed game-theoretic approaches to investigate the strategic interactions between webmasters and search engines. For example, the work of Ben-Basat and colleagues (2021) introduced a formal model wherein webmasters compete for ranking positions under the assumption of a known algorithm. While insightful, such models are often limited by their reliance on the assumption that webmasters possess either complete or partial information about the ranking process. The current study addresses this limitation by introducing a framework in which the algorithm is treated as uncertain or probabilistic.

Other foundational contributions, such as those by Immorlica et al. (2005) and Devanur et al. (2011), have utilized concepts from mechanism design and auction theory to examine bidding behaviors within sponsored search models. These studies have yielded valuable perspectives on strategic decision-making in known algorithmic environments. In contrast, the present research shifts the focus toward organic search rankings and emphasizes the unpredictable nature of the algorithm. This perspective aligns more closely with recent developments in adversarial and robust optimization within the field of operations research.

The issue of algorithmic opacity has also been examined in the context of transparency and fairness. For instance, Hosanagar et al. (2020) highlight the inherent power imbalance that arises from the lack of visibility into search engine operations. Meanwhile, advancements in machine learning have focused on enhancing the interpretability and accountability of ranking systems. However, these efforts have typically prioritized ethical considerations rather than optimization strategies. The novel contribution of this paper lies in modeling the SEO challenge as a Bayesian game. In this framework, each webmaster makes content-related decisions based on probabilistic beliefs about the nature of the search engine’s ranking algorithm. This approach offers a more realistic representation of the environment in which webmasters operate, acknowledging that exact ranking functions are proprietary and subject to frequent change. By employing Markov Decision Processes (MDPs) alongside simulation-based methods for equilibrium computation, this study proposes a robust and adaptable framework that extends many of the simplifying assumptions found in earlier research.

4. Proposed Methodology

Data Collection:

This study uses keyword trend data from Google Trends, a public platform that reflects the relative popularity of search queries over time. Two keywords were selected: Book, which shows relatively stable interest, and Football, which displays clear periodic fluctuations aligned with seasonal events and tournaments. These contrasting patterns allow for evaluating how forecasting models handle trend stability versus volatility and how well they capture uncertainty in different temporal contexts. The dataset covers a five-year period, from May 10, 2020, to May 11, 2025, with weekly observations, resulting in 261 data points per keyword. Popularity values range from 0 to 100, representing normalized interest scores defined by Google Trends. The data was exported in CSV format, including a single timestamp column labeled "Week" and separate columns for each keyword.

On average, Book has a trend rank of 28, while Football scores slightly higher at 30. Although their average popularity is similar, Football shows greater variability and distinct peaks during major sporting events. This contrast provides a meaningful basis for assessing the ability of forecasting models to capture trend dynamics and quantify predictive uncertainty.

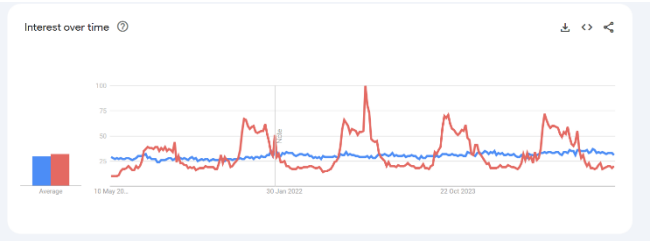


Fig. 1.

Exploratory Data Analysis and Data Processing

An initial exploratory data analysis (EDA) was conducted to examine temporal patterns, trend stability, and lag dependencies within the keyword time series. Visual inspection (Fig. 1) confirmed that Book maintains a relatively steady trend with minor fluctuations, whereas Football exhibits clear periodic spikes linked to seasonal events and tournaments. These contrasting behaviors provide a strong basis for evaluating model performance under both stable and volatile conditions.

The original dataset contained 262 weekly observations. To prepare the data for supervised learning, two lag features were created for each keyword, representing values from one and two weeks prior (lag-1 and lag-2). This approach captures short-term temporal dependencies while keeping the feature space manageable, particularly important given the weekly resolution of the data. The first two observations, which could not accommodate lag values, were excluded, leaving 260 samples for modeling. A chronological 90-10 split was applied to maintain temporal integrity, with 234 points used for training and the remaining 26 reserved for testing. This method prevents data leakage and reflects realistic forecasting conditions.

Descriptive statistics were computed for the full dataset, as well as the training and test subsets (see Fig. 2 and Fig. 3). While average popularity remained consistent, Football demonstrated a wider range, with test data showing a notable decline in peak values. This shift emphasizes the importance of accounting for distributional changes when partitioning time series data. The final input matrix consisted of four lag-based features: book_lag1, book_lag2, football_lag1, and football_lag2, which were used to predict future keyword trends.

	Average	Max	Min
Original	30	38	24
Train	30	38	25
Test	30	37	24

Fig. 2.

	Average	Max	Min
Original	32	100	10
Train	32	100	10
Test	32	73	15

Fig. 3.

Model Selection:

To meet the dual objectives of accurate forecasting and reliable uncertainty quantification, three distinct model classes were employed: Gradient Boosting Regressor (GBR), Random Forest (RF), and a neural-network-based Auxiliary Interval Predictor. These models represent complementary approaches to generating prediction intervals, allowing for a balanced comparison in terms of interpretability, flexibility, and calibration.

The first model, Gradient Boosting Regressor (GBR) with quantile loss, estimates specific quantiles of the target distribution. Separate models were trained for the 5th, 50th, and 95th percentiles to produce lower, median, and upper bounds of the prediction interval. GBR was chosen for its robustness to outliers, ability to model nonlinear relationships, and native support for quantile regression. It was configured with 500 estimators, a maximum depth of 11, minimum leaf size of 9, and a learning rate of 0.05, tuned to balance interval width and coverage.

The second model, Random Forest (RF) Regressor, constructs intervals using ensemble predictions, following Meinshausen's method. Intervals are derived from empirical percentiles across the ensemble. The model includes 100 estimators and a maximum depth of 5, providing a conservative structure that avoids overfitting and yields stable interval estimates. This approach captures model uncertainty through prediction variance and is computationally efficient.

The third model, from IBM's UQ360 toolkit, is an Auxiliary Interval Predictor, which uses a neural network to learn both point forecasts and interval bounds. Implemented as a multi-layer perceptron with 200 hidden units, it was trained over 10 outer loops with 70 main iterations and 20 auxiliary iterations per loop. The model optimizes a composite loss function that balances interval sharpness, coverage, and smoothness. Regularization parameters were set to penalize noisy bounds ($\lambda_noise = 0.007$), enforce consistency ($\lambda_match = 10$), and promote tighter intervals ($\lambda_sharpness = 10$). This method adapts well to local variance and supports calibrated uncertainty estimation.

All models were trained on the same lag-based feature set and evaluated on a common test set to ensure comparability. Performance was assessed using standard accuracy metrics along with prediction interval metrics, including Prediction Interval Coverage Probability (PICP) and Mean Prediction Interval Width (MPIW). This combination of statistical and neural approaches provides a comprehensive view of uncertainty modeling in keyword trend forecasting.

5. Results and discussion:

	Model	Mean Predicted Rank	Max Predicted Rank	Min Predicted Rank	PICP	MPIW
0	Gradient Boosting	32.586862	33.650486	31.241741	0.538462	3.114545
1	Random Forest	32.519461	33.150854	31.872879	0.346154	2.283495
2	Auxiliary Interval	33.846638	37.182354	31.155132	0.692308	3.552052

Fig. 4.

	Model	Mean Predicted Rank	Max Predicted Rank	Min Predicted Rank	PICP	MPIW
0	Gradient Boosting	29.113122	62.598245	16.785503	0.423077	10.398165
1	Random Forest	31.005395	63.740773	17.828495	0.461538	8.819871
2	Auxiliary Interval	37.900616	64.184509	24.480402	0.461538	15.815251

Fig. 5.

Each model produced 90% prediction intervals, evaluated using three key metrics: mean predicted rank, Mean Prediction Interval Width (MPIW), and Prediction Interval Coverage Probability (PICP). Visualizations (Fig. 2–7) further illustrate each model's median forecasts alongside their prediction intervals, plotted against actual observed values to highlight accuracy and uncertainty behavior.

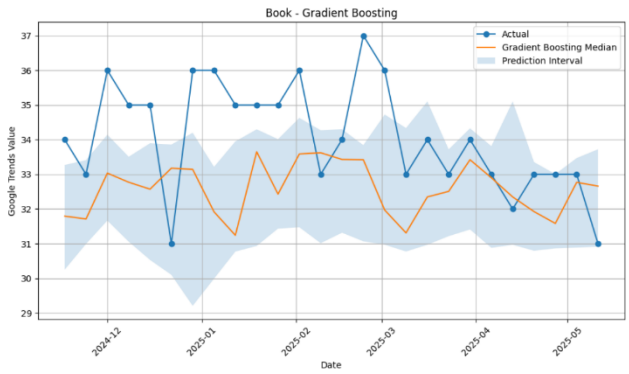
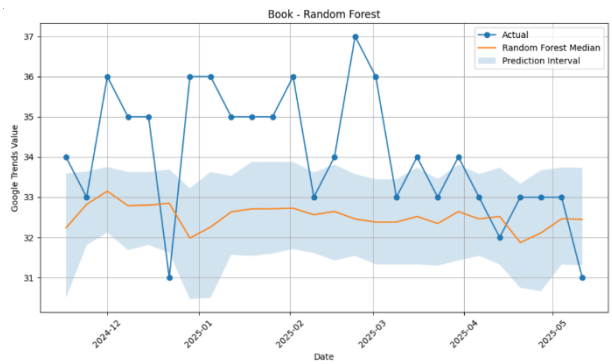


Fig. 6.



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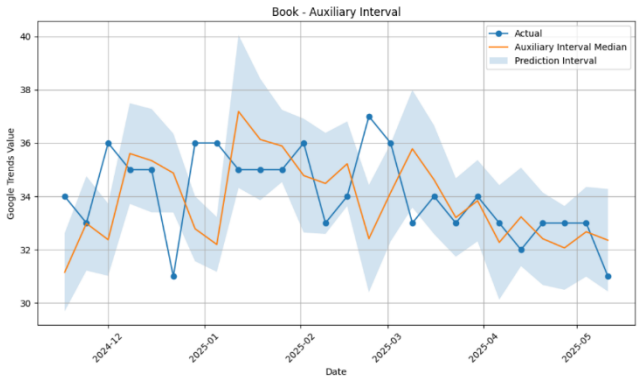

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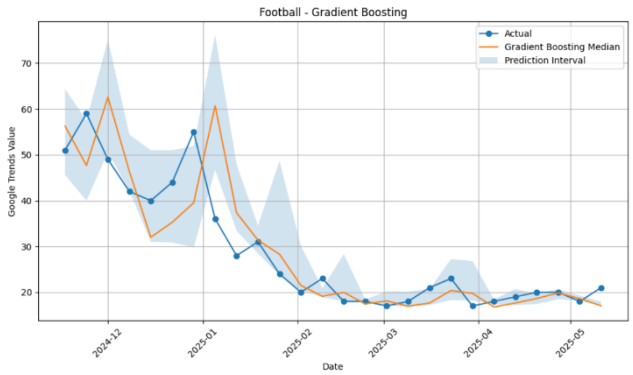

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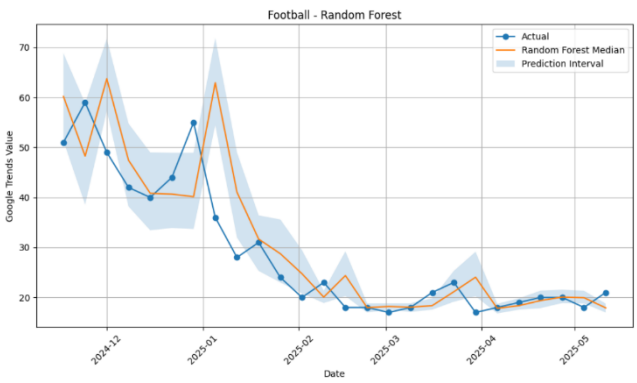

Fig. 10.

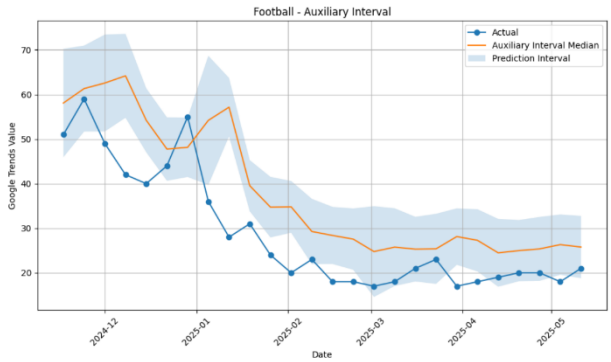

Fig. 11.

A. Book

For the keyword Book, all models produced stable predictions with narrow intervals, reflecting its historically low volatility.

- Gradient Boosting Regressor (GBR) achieved a mean predicted rank of 32.59 (range: 31.24–33.65) and the narrowest interval (MPIW = 3.11), but with a low coverage (PICP = 53.85%), indicating overconfidence.
- Random Forest (RF) yielded the narrowest intervals (MPIW = 2.28) but with the lowest coverage (PICP = 34.62%), severely underestimating uncertainty.
- Auxiliary Interval Predictor achieved the best coverage (PICP = 69.23%) with a moderate interval width (MPIW = 3.55), indicating a better balance between sharpness and reliability.
- Despite suboptimal coverage across models, all effectively reflected the keyword's stability. The Auxiliary model stood out in uncertainty calibration.

B. Football

Forecasting Football was more challenging due to its high volatility and event-driven spikes.

- GBR showed a mean rank of 29.11, wide intervals (MPIW = 10.40), and poor coverage (PICP = 42.31%).
- RF slightly improved coverage (PICP = 46.15%) with narrower intervals (MPIW = 8.82), but still fell short of the 90% target.

- Auxiliary Predictor produced the widest intervals (MPIW = 15.82) but matched RF in coverage (PICP = 46.15%), suggesting difficulty in modeling extreme fluctuations despite higher uncertainty estimation.

Visualizations (Fig. 6–11) confirmed these findings. While no model achieved ideal coverage, their intervals reflected meaningful differences in trend volatility. This distinction is valuable in applications like SEO automation or AI-driven systems, where conveying uncertainty is critical.

Instead of transmitting full intervals—which can be costly in LLM-based pipelines—a practical solution is to report summary metrics like MPIW and PICP, offering interpretable signals of forecast reliability with minimal overhead.

5. Conclusion

Forecasting keyword trends, especially for volatile terms like *Football*, is inherently challenging. While none of the models achieved ideal 90% coverage, they effectively reflected keyword stability through interval width and coverage probability.

Key takeaways from the study include:

- Prediction intervals matter more than point forecasts for high-volatility keywords.
- Auxiliary Interval Predictor offered the best uncertainty calibration, especially for stable keywords like *Book*.
- MPIW and PICP are practical metrics to convey forecast confidence in automated systems.
- Future improvements should include hybrid models and integration of external signals (e.g., event data).

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